Hybrid Rag Solution

SECTION 03 / GENSTOCKFOLIO

Generation pipeline

Retrieved evidence, grounded answers, and source citations.

LangChain

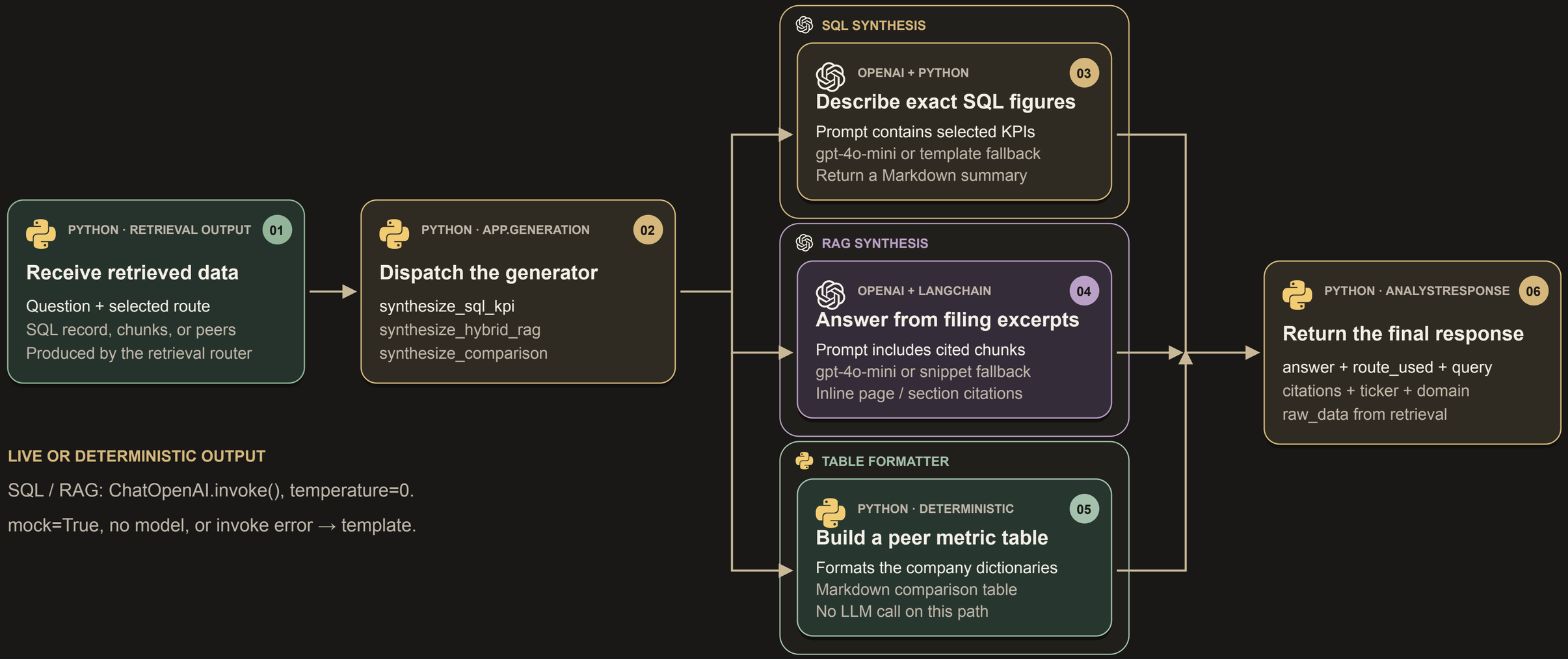
OpenAI

Python

02 DIAGRAMS / PIPELINE WALKTHROUGH

From retrieved evidence to an answer

Each retrieval route selects its own synthesis function; peer comparison always uses a deterministic table.



LIVE OR DETERMINISTIC OUTPUT

SQL / RAG: ChatOpenAI.invoke(), temperature=0.

mock=True, no model, or invoke error → template.

The functions return answer strings. The retrieval router adds metadata; no token streaming or citation validator runs here.

RAG generation: keep evidence in the prompt

The live narrative path uses retrieved excerpts and inline citation labels to guide a grounded answer.

PROMPT & MODEL

ANSWER & RESPONSE

PYTHON · RETRIEVAL 01

Receive cited chunks

query + retrieved chunk payloads
text, page, domain, citation
Empty input → no-evidence notice

LANGCHAIN · MESSAGES 02

Assemble the prompt

Question + Source [citation]
Join excerpts with separators
Use ONLY supplied evidence

OPENAI 03

Generate the answer

ChatOpenAI: gpt-4o-mini
temperature=0.0; invoke()
System + Human messages

PYTHON · GENERATION 04

Return model content

resp.content becomes answer
No post-generation validation
Complete string, no SSE tokens

PYTHON · ROUTER 05

Package the response

Router adds the retrieved chunks
route_used = hybrid_rag
AnalystResponse to Python caller

CITATION INSTRUCTION

Every key factual claim should cite its excerpt:
[Page - X, Section - Y]
The prompt requests this format; code does not verify
citation correctness or detect unsupported claims.

DETERMINISTIC FALLBACK

Used for mock mode, missing model, or invoke failure.
Format the first 280 characters of each excerpt
as a bullet, retaining its citation label.
This is excerpt formatting, not a live LLM answer.

The comparison path bypasses this LLM flow entirely. SQL synthesis uses a separate exact-figures prompt.

The backend result is available through Python / CLI; Flutter's current chat widget is not connected to these functions.